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European Journal of Neurology 2006, **13:** 1112–1117 doi:10.1111/j.1468-1331.2006.01423.x

Bowel dysfunction in patients with cauda equina lesions

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Keywords:
anal incontinence, bowel, cauda equina, constipation, conus medullaris, external anal sphincter

Received 19 August 2005
Accepted 19 October 2005

Despite their serious sequels on bowel function, lesions of the cauda equina have not been previously systematically studied in larger patient populations. This was the aim of the present report. From the registrars of a diagnostic and rehabilitation centres 67 patients with clinical, electrodiagnostic and radiological findings supportive of the cauda equina lesions were recruited. The Slovene versions of the standard questionnaires for anal incontinence and constipation were used. The responses were scored, impairments categorized and previous treatments noted. Neurological examination, electromyography (EMG) of lumbo-sacral myotomes, quantitative anal sphincter EMG and electromyographic evaluation of the sacral reflex were performed. Severe anal incontinence/constipation was reported by 18%/0%, moderate by 36%/33%, and slight by 28%/43% of our patients. Twenty-one per cent of patients wore pads continuously and 14% occasionally. More than half of the patients (60%) reported changes in their lifestyle due to anal incontinence. No patient had completely normal findings on neurological examination. Perianal sensory loss correlated ($P < 0.05$) with anal incontinence and gender with constipation (women >men). Only two patients had received medical attention for bowel dysfunction. Study thus demonstrated significant bowel impairment in patients with lesions of the cauda equina, which has received insufficient medical attention.

Introduction

Neurogenic lesions are an important cause of bowel dysfunction, and large populations of patients with strokes [1], multiple sclerosis [2], Parkinson's disease [3], and long-term spinal cord injuries [4,5] have been reported. Although cauda equina or conus medullaris lesions also have serious consequences on bowel function, there is little data available from systematically studied populations of these patients [6].

For cauda equina and conus medullaris lesions, annual incidence rate of 0.49 and period prevalence of 14.8 patients per 100 000 people were estimated (S. Podnar, unpublished data). The exact distinction between cauda equina and conus medullaris lesions is difficult and in the present study, the term 'cauda equina lesions' has been used disregarding the exact localization of the lesion.

In the present study, the bowel dysfunction of patients with long-standing lesions of the cauda equina has been reported, and correlated it with findings on clinical neurological examination and quantitative electromyography (EMG) in the external anal sphincter (EAS) muscles. Bladder dysfunction [7] and male sexual dysfunction [8] in the same population were reported elsewhere.

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Materials and methods

Subjects

Eighty-eight patients from the databases of the main diagnostic and rehabilitation centres in Slovenia were invited to participate in the study. The patients were diagnosed to have cauda equina lesion by history pointing to an adequate causative event, demonstration either perianal sensory loss or spontaneous denervation activity in lower spinal myotomes on early EMG examination (performed 3–10 weeks after the onset of the condition), and by radiological findings supportive of such injury. The National Ethics Committee of Slovenia approved the study, and all subjects provided informed consent.

Bowel (dys)function questionnaires

At the clinical and EMG examination, all patients were asked about bowel function. Afterwards, Slovene versions of the anal incontinence [9] and chronic constipation [10] questionnaires were sent to all patients. These questionnaires include 18 questions covering five aspects of bowel (dys)function (Tables 1 and 2):

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consistency of faeces (Q1), anal incontinence (six questions: Q2–7), sensation during defecation (Q8), pain and other associated problems (Q9), bowel and general medical conditions (Q10), and constipation (eight questions: Q11–18). Questions were scored 0–1 (Q8), 0–2 (Q16), 0–3 (Q2), 0–4 (Q3–7, Q11–15, Q17–18) (Tables 1 and 2) [9,10]. More pronounced bowel dysfunction scored higher. Mild anal incontinence was defined as incontinence of flatus only (anal incontinence score: 1–5), moderate when incontinence of liquid stools occurred a few times a week (score: 6–10), and severe when daily incontinence of liquid or solid stools was present (score: 11–20). Perfect anal continence scored 0. Patient constipation was defined as absent (constipation scores: 0–4) with no evacuatory problems, mild (score: 5–10) with rare and mild problems, moderate (score: 11–20) with more pronounced and frequent troubles, or severe (score: 21–30) with continuous and very troublesome defecation [10].

The number of patients evaluated for their bowel disorder by a colorectal surgeon or gastroenterologist was noted.

Neurological and electrodiagnostic evaluation

In all patients, neurological examination of the trunk and lower limbs was performed. General scores of clinical findings were determined by summation of scores from both sides [8].

Electromyography was performed using a standard concentric EMG needle, and an advanced EMG system (Keypoint; Medtronic Functional Diagnostics, Skovlunde, Denmark) with standard settings (filters: 5 Hz–10 kHz). Several muscles belonging to the lumbo-sacral myotomes were examined. This included observation of needle insertion and spontaneous denervation activity, the characteristics of motor unit potentials (MUPs), and assessment of the interference pattern at several levels of voluntary activation. In addition, in the EAS muscles a precisely standardized quantitative EMG examination protocol was followed [11]. Protocol includes counting the number of continuously firing MUPs during relaxation [12], observing the increase in MUP firing to perianal touch (i.e. sacral reflex activation), and sampling of MUPs by multi-MUP analysis. MUP parameters of area, duration, and number of turns [13] were compared with normative data for mean values and 'outliers' [14]. The EMG findings were graded into five categories: normal findings (category I), degrees of partial EAS lesions (categories II–IV: II slight; III moderate; IV severe), and complete denervation of all EAS muscles bilaterally (category V) [7,8].

The effect of age, gender, aetiology, duration and level of the neural lesion, perianal sensation, and the grade of EMG abnormality on anal incontinence and constipation, were studied using multiple linear regression analysis by MATLAB (The MathWorks, Natick, MA, USA). In addition, correlation between perianal sensation and sensation during defecation, as well as correlation between anal incontinence and constipation scores were calculated by the non-parametric Spearman correlation coefficient.

Table 1 Symptoms of anal incontinence in 67 patients with chronic lesions of the cauda equina

Frequency	Flatus (%)	Liquid stools (%)	Solid stools (%)	Pads (%)	Lifestyle (%)
Never	20	47	56	65	52
Rarely	6	11	8	2	8
Sometimes	24	17	15	8	17
Usually	21	12	11	5	12
Always	29	14	11	21	12

Rarely = < 1/month; sometimes = < 1/week, ≥1/month; usually = < 1/day ≥1/week; always = ≥1/day [9].

Results

A group of 67 patients (48 men) was recruited. The median age was 47 years (range: 20–78), and the

Table 2 Responses of 51 patients with chronic lesions of the cauda equina to questions used in the constipation scoring system [10]

Score	Frequency (%)	Difficulty (%)	Completeness (%)	Pain (%)	Time (%)	Failure (%)	History (%)	Assistance (%)
0	59	33	28	37	17	57	42	52
1	14	24	15	20	37	29	42	15
2	22	24	19	20	25	6	6	33
3	5	12	15	18	10	2	2	–
4	–	8	23	6	12	6	8	–

Frequency of bowel movements: 0 = 1–2 per 1–2 days, 1 = 2 per week, 2 = 1 per week, 3 = < 1 per week; frequency of difficulty (painful evacuation effort), completeness (feeling of incomplete evacuation), pain (abdominal pain): 0 = never, 1 = rarely, 2 = sometimes, 3 = usually, 4 = always; time (minutes in the lavatory): 0 = < 5, 1 = 5–10, 2 = 10–20, 3 = 20–30, 4 = > 40; failure (unsuccessful evacuations per 24 h): 0 = never, 1 = 1–3, 2 = 3–6; history (years of constipation): 0 = 0, 1 = 1–5, 2 = 5–10, 3 = 10–20, 4 = > 20; assistance (type of assistance): 0 = without assistance, 1 = stimulative laxatives, 2 = digital assistance or enema [10].

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median time since the lesion to the cauda equina was 21 months (range: 5–240).

Data obtained by questionnaires

The completed bowel dysfunction questionnaires were received from 51 patients. In the remaining 16 patients, data obtained during clinical and EMG examination enabled us to calculate the anal incontinence scores, but not constipation scores. Responses to principal questions regarding anal incontinence are presented in Table 1. Six patients reporting incontinence of solid stools never had liquid stools. Fifteen of 26 patients without faecal incontinence reported incontinence of flatus. No sensation during defecation was reported by 39% patients. The anal incontinence score revealed no impairment in 18% patients, mild impairment in 28% patients, moderate impairment in 36% patients, and severe impairment in 18% patients (median score: 6/20, range: 0–15).

Constipation was reported by 68% of men and 93% of women. Responses to questions regarding constipation are presented in Table 2. The frequency of bowel evacuation varied in our patients from one per 3 weeks to five per day. Many patients reported a concomitant evacuation of small quantities of dry faeces during almost each voiding (typically performed in a sitting position using abdominal muscle contraction). Seventeen of our patients regularly evacuated their bowels with the manual assistance or enema (mainly the former). The constipation score revealed no impairment in 25%, mild impairment in 42% and moderate impairment in 33% of patients, whilst none of our patients had severe impairment according to the questionnaire applied (median score: 6/30, range: 0–20).

Only eight of 51 patients, who completed bowel dysfunction questionnaires, reported no associated symptoms. The most frequently reported associated symptoms were: excessive straining on defecation (24 patients), rectal pain (16 patients), mucous discharge (13 patients), abdominal cramps (12 patients), bloating (11 patients), rectal bleeding (nine patients) and a sensation of rectal prolapse (eight patients). In addition, 18 patients had haemorrhoids and one an anal fissure. Severely reduced or absent sensation during defecation was reported by 39% of 61 patients. Hard and dry faeces reported 37%, hard and formed 26%, soft and formed 28%, and liquid or semi-liquid stools 9% of 55 patients.

One patient was totally, and another partially wheelchair bound. The remaining patients were ambulant, some with varying degrees of lower limb paresis. Uni- or bilateral plantar flexion weakness was particularly common. Before inclusion into our study only two of our patients had been evaluated by colorectal surgeon or gastroenterologist, and none had surgery for bowel dysfunction.

Findings of neurological and electrodiagnostic examinations (Table 3)

No patient had completely normal findings, and none had upper motor neurone signs on clinical examination. The MUP count during relaxation was below the normative range in at least one EAS muscle in 53% of 64 patients. Sacral reflex activation was normal in only 16% of 64 patients. EMG findings were normal (category I) in 12% patients and abnormal (categories II–V) in 88% of 65 patients. No needle insertion activity or only spontaneous denervation activity (with no MUP recruitment) could be detected in either the subcutaneous or the deeper EAS muscles bilaterally in 8% patients (complete muscle denervation).

Table 3 Findings of neurological examination and electrodiagnostic studies in 67 patients with chronic lesions of the cauda equina or the conus medullaris

Findings	Frequency (%)
Neurological examination	
Knee flexion	
Normal	51
Slightly reduced	38
Moderately reduced	9
Severely reduced	2
Perianal sensation	
Normal	4
Slightly reduced	29
Moderately reduced	21
Severely reduced	25
Absent	21
Ankle jerks	
Normal	14
Reduced	49
Non-elicitable	37
Electrodiagnostic studies	
MUP count during relaxation	
Normal	47
Reduced	53
Sacral reflex activation	
Normal	16
Reduced	53
Absent	31
Multi-MUP analysis	
Category I	12
Category II	12
Category III	49
Category IV	19
Category V	8

Category I = normal, categories II–IV = II slight; III moderate; IV severe partial neuropathic lesions, category V = complete denervation [7,8].

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Table 4 Correlation of clinical (perianal sensation) and electromyographic (EMG) findings with median anal incontinence and constipation scores in 51 patients with chronic lesions of the cauda equina or the conus medullaris

	Anal incontinence (0–20)	Constipation (0–30)
Perianal sensation		
Normal	0	3
Mildly reduced	3	4
Moderately reduced	6	8
Severely reduced/absent	10	10
EMG findings		
Category I	4	9
Category II	3	4
Category III	4.5	8
Categories IV–V	8	7

EMG findings: category I = normal, categories II–IV = II slight; III moderate; IV severe partial neuropathic lesions, category V = complete denervation [7,8].

Correlation of neurological and electrodiagnostic findings with bowel function (Table 4)

Using multiple linear regression analysis, anal incontinence was found to be significantly associated with diminished perianal sensation ($P = 0.0002$). Other independent variables included into our regression analysis (age, gender, level, aetiology and duration of the neural lesion, and the grade of EMG abnormality) did not have significant effect ($P > 0.05$) on anal incontinence. Factors included into our regression equation explained 40% of variability of dependent variable (anal incontinence, $R^2 = 0.40$).

Constipation was significantly associated only with female gender ($P < 0.03$). However, independent variables included into this regression analysis explained only 20% of variability of dependent variable (constipation, $R^2 = 0.20$).

Using the Spearman correlation coefficient sensation during defecation was strongly (Spearman $r = 0.529$; 95% CI: 0.326–0.686) and significantly ($P < 0.0001$) correlated with perianal sensation. Anal incontinence and constipation were also significantly ($P < 0.05$) correlated (Spearman $r = 0.246$; 95% CI: −0.002–0.464); concomitant faecal incontinence (score > 5) and constipation (score > 4) was reported by 51% of 51 patients.

Discussion

The present study revealed important anal incontinence and chronic constipation in population of patients with cauda equina lesions. Although not formally evaluated in a questionnaire, bowel dysfunction seemed to be more troublesome than bladder or sexual problems in our chronic patients. In addition, perianal sensory deficit and EMG abnormalities were also found in majority of patients. The cauda equina conveys motor nerve fibres innervating lower sacral myotomes, sensory fibres from the sacral dermatomes, and sacral parasympathetic fibres, so these abnormalities are not unexpected.

Our patients with cauda equina lesions reported a much higher rate of faecal incontinence than the general population. Within the general population 1.4% of people older than 40 years soil underwear at least several times a month [15]. Furthermore, most of our patients related the onset of bowel dysfunction to the neural lesion. Our results (Table 1) also show more severe faecal incontinence to that found in a previous study (see also constipation below), in which 30% of cauda equina patients 'were occasionally frankly incontinent for stools' [6]. As this study did not use the same questionnaire, comparison with our results is difficult. Frequency of anal incontinence found in our study was similar to findings in patients with multiple sclerosis [2], and after spinal cord injuries [4], but higher than in stroke patients [1]. The main difference, however, between our patients and these three patient populations was in the relative importance of anal incontinence in the patient's total disability. Only two of our patients were not ambulant, which contrasts much higher frequency of disturbed gait and other disabilities in the other three neurological conditions causing bowel dysfunction.

The essential prerequisite for anal continence is higher pressure within the anal canal compared with the pressure within the rectum. The internal anal sphincter muscle, which provides most of the resting pressure within the anal canal, is driven by sympathetic outflow [16], which bypasses the cauda equina. To preserve continence during increases of abdominal pressure and during anal sampling (recto-anal inhibitory reflex), additional quick contraction of the EAS and pelvic floor muscles is needed. For quick reflex responses, apart from competent muscles, normal anorectal sensory function is also essential. Motor fibres that innervate the EAS and pelvic floor muscles, and probably also sensory fibres that convey anorectal sensation (see below) travel mainly within the cauda equina.

Constipation constituted additional bowel impairment in our population. Constipation was again much more prevalent than in the general adult population, in which 2–9% are affected [2, 4, 17, 18]. Almost all women (93%) were constipated, and they were more severely affected than men, which is in accordance with findings in the general population [17]. Severity of constipation in our population was similar to that in a

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previous study of patients with lesions of the cauda equina (72%). However, an exact comparison is difficult because of the insufficient data provided by that study [6]. The frequency was also similar to that reported in patients with multiple sclerosis [2], and in Parkinson's disease [3]. In contrast, the prevalence was much lower after spinal cord injury (17%) [4]. Although most of our patients reported constipation, none of them had severe constipation according to the grading system applied (Table 2). This is unexpected because, except patients with daily faecal incontinence, constipation caused more inconvenience than other sacral dysfunction to most of them. None of questionnaires used in this study was validated in patients with neurogenic lesions. Lower scores were probably obtained due to the short duration of constipation (time interval since neural injury), sensory loss (with inability to feel pain, and incompleteness of bowel emptying), and using a routine (47% of our patients required assistance for defecation) that increased the frequency of defecation, reduced the usual time spent on the lavatory and reduced the number of defecation failures. A common strategy used by a number of patients was to take laxatives 1–2 times a week when they planned to stay at home. They are confined to the lavatory for that day, and they succeeded in emptying their bowels, which also relieves incontinence (when present) for the next couple of days. Many patients also evacuate bowels manually, which they consider a major inconvenience. The severity of constipation is also illustrated by high frequency of symptoms associated with impaired defecation.

In patients with cauda equina lesions constipation is probably caused by several mechanisms. One is slow colon transit caused by parasympathetic denervation [19,20]. The left colon and rectosigma are supplied by sacral (S2–S4) parasympathetic fibres. Chronic rectal distention may additionally slow down the colonic transit. Filling of the rectum normally produces the desire to defecate. This depends on intact sensory pathways that (probably) include the cauda equina (see below). Constipation (and chronic rectal distention) may, furthermore, also result from defective evacuation. A number of our patients who assist defecation manually point to pelvic floor dysfunction as an additional factor.

Although different populations of nerve fibres mediate perianal sensation and anal continence, as well as propulsion of faeces in the colon, they were all well correlated. Sensory loss in the lower sacral dermatomes is thus a good clinical parameter for the assessment of patients with cauda equina lesions (Table 4). However, three of 67 patients included in this study, who definitely had a cauda equina lesion, did not have any perianal sensory deficit. One of these patients had completely normal bowel function, whilst the other two had mild anal incontinence, in one accompanied by mild constipation. Perianal sensory loss was also highly correlated with lost sensation during defecation. This is in accordance with the view that rectal fullness is perceived mainly by proprioceptors of the pelvic floor muscles [21] that accompany somatic motor fibres on their way through the cauda equina, and not by afferent fibres that accompany sympathetic fibres and bypass the cauda equina.

To test the sacral segment motor fibres, quantitative EMG of the EAS muscle was performed. The MUP count during relaxation (provides a crude estimate of the number of motor units remaining after the neural lesion and of segmental sensory input) [12], and analysis of MUP parameters (provides information on the EAS muscle reinnervation) did not correlate with bowel function. This is probably because of high re-innervation ability of the peripheral motor axon, which contrasts with poor ability for repair of the central sensory axon and pre-ganglionic parasympathetic axon. This might reduce the value of EMG for assessment of the initial severity of lesions in chronic situation. In contrast, EMG in the early phase after cauda equina lesions is more useful, because assessment of spontaneous denervation activity has high sensitivity for motor axon damage. However, in individual patients with unclear diagnosis EMG performed in the chronic phase can provide important data.

It seems that our patients were inadequately managed for their bowel disorders, as only 4% patients with anorectal dysfunction were examined by a colorectal surgeon or a gastroenterologist before inclusion into the study. In my opinion, to optimize their management, every patient with a cauda equina lesion and bowel dysfunction should be seen by a physician with expertise in treating bowel dysfunction in neurogenic patients, as judiously applied measures (e.g. use of bulking agents, irrigation, enemas, etc.) might greatly improve quality of their life.

Acknowledgements

The study was supported by the Republic of Slovenia Research Agency, grant No. J3 7899. The author thanks Dr Aleš Pražnikar for help with recruitment of patients, Mićo Mrkaić and Nacek Zidar for statistical analysis, and Dr Dianne Jones for language review.

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